

$$3-c) [(p \vee q) \wedge (q \vee p)] \Rightarrow (p \vee q)$$

0	0	0	0	0	0	0	1	0	0	0
0	1	1	1	1	1	0	1	0	1	1
1	1	0	1	0	1	1	1	1	0	1
1	1	1	1	1	1	1	1	1	1	1

tautology

$$d) [p \wedge (\neg p)] \Rightarrow q$$

0	0	1	1	0
0	0	1	1	1
1	0	0	1	0
1	0	0	1	1

tautology

$$f) \{ [p \wedge (\neg q)] \Rightarrow r \} \wedge [(p \vee q) \Rightarrow r]$$

0	0	1	1	0	1
0	0	0	1	0	0
1	1	1	0	0	0
1	0	0	1	0	0
0	0	1	1	1	1
0	0	0	1	1	1
1	1	1	1	1	1
1	0	0	1	1	1

0	0	0	1	0
0	1	1	0	0
1	1	0	0	0
1	1	1	0	0
0	0	0	1	1
0	1	1	1	1
1	1	0	1	1
1	1	1	1	1

tautology

$$10- v[q \vee s] = 1$$

$$v(r) = 0$$

$$v[\neg p \Rightarrow q] = 0$$

$$v(q) = 0$$

$$v[\neg p \Rightarrow (q \wedge \neg r)] = 0$$

$$v(s) = 1$$

$$v(p) = 0$$

$$11) [(q \Rightarrow \neg r) \wedge (\neg q \vee s)] =$$

